LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 12 Magnetic Effects of Electric Current

Chapter / Topic: Magnetic Effects of Electric Current · Week 6 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 25 minutes each: Day 1 on the day you study the chapter, then Day 2 and Day 3. Write answers in your notebook and check them with the key the same day. Copy every mistake into your error log.
2. For every direction question, use your hand and say the rule out loud. Right-hand thumb rule for the field; Fleming's left-hand rule for the force.

DAY 1: MAGNETIC FIELDS AND FIELD LINES

Q1. A compass needle is a small:

(a) bar magnet (b) piece of copper
(c) solenoid (d) electric cell

Q2. Around a bar magnet, the magnetic field is strongest:

(a) at the middle of the magnet (b) far away from the magnet
(c) near the poles (d) equally everywhere

Q3. The magnetic field produced by a straight current-carrying wire at a given point becomes stronger if:

(a) the point is moved farther from the wire
(b) the current in the wire is increased
(c) the wire is painted
(d) the current is reduced

Q4. What did Hans Christian Oersted observe in 1820, and why was it important?

Q5. How does a field-line diagram show where a magnetic field is strong and where it is weak?

Q6. A straight vertical wire passes through a horizontal sheet of cardboard and carries current upwards. Describe the field lines on the cardboard, and give their direction as seen from above. What changes if the current is reversed?

Q7. Why is the magnetic field at the centre of a circular coil of 50 turns about 50 times as strong as that of a single turn carrying the same current?

DAY 2: FORCE ON A CONDUCTOR AND ELECTROMAGNETS

Q8. In Fleming's left-hand rule, the thumb points in the direction of the:

(a) force (motion) on the conductor
(b) current
(c) magnetic field
(d) north pole

Q9. The force on a current-carrying conductor in a magnetic field is greatest when the angle between the current and the field is:

(a) 0° (b) 30°
(c) 45° (d) 90°

Q10. A beam of electrons moves from left to right across this page. A magnetic field acts into the page. The force on the beam is towards the:

(a) top of the page (b) bottom of the page
(c) inside of the page (d) outside of the page

Q11. How can you find out which end of a current-carrying solenoid is its north pole, using only a compass?

Q12. Why is soft iron, and not steel, used as the core of an electromagnet?

Q13. A copper rod PQ is hung between the poles of a horseshoe magnet. When current flows from P to Q, the rod moves to the left. What happens if: (a) the current is reversed? (b) the poles of the magnet are swapped? (c) a larger current is passed?

Q14. Give three advantages of an electromagnet over a permanent magnet.

DAY 3: DOMESTIC CIRCUITS, MIXED PRACTICE AND CHALLENGE

Q15. **Assertion (A):** In a domestic circuit, the fuse is connected in the neutral wire.
Reason (R): A fuse protects appliances by melting when the current becomes too large.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q16. Short-circuiting happens when:

(a) too many appliances are run from one socket
(b) the earth wire is disconnected
(c) the live wire and the neutral wire come into direct contact
(d) a fuse melts

Q17. In India, the potential difference between the live wire and the neutral wire of the household supply is:

(a) 12 V (b) 110 V
(c) 220 V (d) 440 V

Q18. In a hostel room, a 1320 W room heater, a 440 W iron and a 110 W fan are all plugged into one extension board on a 5 A socket. The supply is 220 V.

(i) Find the total current drawn when all three are on.

(ii) Is this safe on a 5 A socket? What is this situation called?

(iii) Which of the three appliances could safely be used alone on the 5 A socket?

Q19. (a) State the right-hand thumb rule and Fleming's left-hand rule.
(b) A straight wire lies north–south, just above a compass needle, and carries current from south to north. Which way does the north pole of the needle turn?
(c) Why does the needle swing back when the current is switched off?

Q20. Someone replaces a blown fuse with a much thicker piece of wire 'so that it never blows again'. Why is this dangerous?

~ Week 6 done after this one. All 13 chapters practised. Superb! ~